

SUPPORTING INFORMATION

Li⁺ extraction from aqueous medium using tetracyanoborate ionic liquids – experiments and ePC- SAFT modeling

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Table S1: Solubility of [emim]⁺ in water at 1 bar and 294.15K (modeled and experimental). ePC-SAFT advanced calculations using the parameters from Tables 2-4. Standard uncertainty is $u(T) = 0.5 \text{ K}$; $u(m) = 0.0001 \text{ g}$; $u(\text{UV-VIS}) = 0.0074 \text{ g g}^{-1}$. The combined expanded uncertainty is $U = 0.0162$ with a coverage factor of $k = 2$ (95% confidence interval).

[emim][tcb]		
$W_{\text{LiCl}_2\text{Feed}}$	$W_{[\text{emim}^+],\text{aq,exp}}$	$W_{[\text{emim}^+],\text{aq,calc.}}$
0	0.0160	0.0160
0.0087	0.0173	0.0173
0.0317	0.0233	0.0232

ARD = 0.1302 %, AAD = $6.5 \cdot 10^{-5}$

Table S2: Solubility of [emim][tcb] in water in comparison with literature data.

$W_{[\text{emim}][\text{tcb}]_{\text{aq}}}$	Temperature / K	Pressure / mPa	Reference
0.0420	296.65	0.1	[1]
0.0325	294.15	0.1	This work.

Table S3: Solubility of [bmim]⁺ in water at 1 bar and 294.15K (modeled and experimental). ePC-SAFT advanced calculations using the parameters from Tables 2-4. Standard uncertainty is $u(T) = 0.5 \text{ K}$; $u(m) = 0.0001 \text{ g}$; $u(\text{UV-VIS}) = 0.017 \text{ g g}^{-1}$. The combined expanded uncertainty is $U = 0.0363$ with a coverage factor of $k = 2$ (95% confidence interval).

[bmim][tcb]		
$W_{\text{LiCl}_2\text{Feed}}$	$W_{[\text{bmim}^+],\text{aq,exp}}$	$W_{[\text{bmim}^+],\text{aq,calc.}}$
0	0.0007	0.0007
0.00246	0.0007	0.0007
0.00859	0.0009	0.0008
0.02087	0.0009	0.0009
0.03216	0.0010	0.0010

ARD = 0.9835 %, AAD = $4.2 \cdot 10^{-5}$

Table S4: Solubility of [hmim]⁺ in water at 1 bar and 294.15K (modeled and experimental). ePC-SAFT advanced calculations using the parameters from Tables 2-4. Standard uncertainty is $u(T) = 0.5 \text{ K}$; $u(m) = 0.0001 \text{ g}$; $u(\text{UV-VIS}) = 0.027 \text{ g g}^{-1}$. The combined expanded uncertainty is $U = 0.0574$ with a coverage factor of $k = 2$ (95% confidence interval).

[hmim][tcb]		
$W_{\text{LiCl}_2\text{Feed}}$	$W_{[\text{hmim}^+],\text{aq,exp}}$	$W_{[\text{hmim}^+],\text{aq,calc.}}$
0	0.0054	0.0054
0.00151	0.0054	0.0054
0.00366	0.0052	0.0055
0.00960	0.0058	0.0055
0.01510	0.0051	0.0055
0.02326	0.0054	0.0056
0.03073	0.0055	0.0056

ARD = 0.2938 %, AAD = $1.8 \cdot 10^{-4}$

Table S5: Solubility of [hmim][tcb] in water in comparison with literature data.

$W_{[hmim][tcb]_{aq}}$	Temperature / K	Pressure / mPa	Reference
0	308.15	0.1	[2]
0.00908	294.15	0.1	This work.

Table S6: Solubility of TBP in water at 1 bar and 294.15K (modeled and experimental) using the parameters from Tables 2-4. Standard uncertainty is $u(T) = 0.5$ K; $u(m) = 0.0001$ g. The combined expanded uncertainty is $U = 0.039$ with a coverage factor of $k=2$ (95% confidence interval).

$W_{LiCl,Feed}$	$x_{TBP,org,exp}$	$x_{TBP,org,calc.}$
0	0.5134	0.5130
0.0085	0.4251	0.4875
0.0323	0.4670	0.4750

ARD = 1.5126 %, AAD = 0.0236

Table S7: Solubility of TBP in water in comparison with literature data.

$x_{TBP,org,exp}$	Temperature / K	Pressure / mPa	Reference
0.493	298.15	0.1	[3]
0.665	303.15	0.1	[4]
0.492	298.15	0.1	[5]
0.563	303.2	0.1	[6]
0.513	294.15	0.1	This work

Table S8: Effects of different ILs on extraction efficiency E_{Li^+} with TBP and ionic liquid as extractant ($w_{Li^+} = 2000$ ppm), $R^w(\frac{O}{A}) = 1$, $T=294.15$ K, $p=1.013$ bar, mixing time 2 hours at 2500 rpm. Standard uncertainty is $u(T) = 0.5$ K; $u(m)=0.0001$ g. The combined expanded uncertainty is $U = 0.001$ with a coverage factor of $k=2$ (95% confidence interval).

IL	$E_{Li^+} / \%$
[emim][tcb]	48.06 (± 10.96)
[bmim][tcb]	65.49 (± 4.57)
[hmim][tcb]	88.64 (± 0.44)

Table S9: Effect of Li^+ content in aqueous feed on extraction efficiency E_{Li} with [emim][tcb] and TBP ($w_{TBP,organic} = 0.80$), $R^w(\frac{O}{A}) = 1$, $T=294.15$ K, $p=1.013$ bar, mixing time 2 hours at 2500 rpm. Standard uncertainty is $u(T) = 0.5$ K; $u(m)=0.0001$ g. The combined expanded uncertainty is $U = 0.0067$ with a coverage factor of $k=2$ (95% confidence interval).

$w_{Li^+,Feed}$	$E_{Li^+} / \%$
0.21	98.49 ± 0.08
197	97.98 ± 0.11
737	95.2 ± 0.04
983	90.43 ± 0.34
1228	90.10 ± 2.49
1474	89.55 ± 0.34
1705	89.17 ± 0.31
1970	85.63 ± 0.04
2193	79.80 ± 1.87
2436	82.52 ± 4.82
2680	83.17 ± 0.51
2923	69.34 ± 4.70

Table S10: Effects of TBP content in the organic phase on extraction efficiency E_{Li^+} with [emim][tcb] and TBP ($w_{TBP,organic} = 0.80$) as extractant ($w_{Li^+} = 2000$ ppm), $R^w(\frac{O}{A}) = 1$, $T=294.15$ K, $p=1.013$ bar, mixing time 2 hours at 2500 rpm. Standard uncertainty is $u(T) = 0.5$ K; $u(m) = 0.0001$ g. The combined expanded uncertainty is $U = 0.006$ with a coverage factor of $k = 2$ (95% confidence interval).

$w_{TBP,org.Phase}$	$E_{Li^+} / \%$
0.60	74.08 (± 5.85)
0.80	88.64 (± 0.44)
0.85	91.39 (± 1.53)
0.88	89.11 (± 1.75)
0.95	70.21 (± 0.93)

Table S11: Effect of $R^w(O/A)$ on the extraction efficiency E^{Li^+} with [emim][tcb] and TBP as extractant ($w_{TBP,organic} = 0.85$, $w_{Li^+} = 2000$ ppm), $T=294.15$ K, $p=1.013$ bar, mixing time 2 hours at 2500 rpm. ePC-SAFT advanced calculations using the parameters from Tables 2-4. Standard uncertainty is $u(T) = 0.5$ K; $u(m) = 0.0001$ g. The combined expanded uncertainty is $U = 0.006$ with a coverage factor of $k = 2$ (95% confidence interval).

$R^w(\frac{O}{A})$	$E_{Li^+,exp} / \%$	$E_{Li^+,calc} / \%$
0.522	75.39	78.16
0.985	89.01	88.58
1.976	93.50	95.02
3.015	96.81	97.05
3.933	98.33	97.04

ARD = 0.1007 %, AAD = 1.2510

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